

STA642 Mega For Final Term March 2022

Prepared by Everblue

Lec (22-45), Topic (108-185)

Objective (11 files)

- Sta642 imP fOr MuLaS pg7
- Sta642 formula pg8
- Sta642 formulas pg10
- STA642-formulas pg4
- STA 642 random formulas pg10
- STA642 current paper mcqs 15march2022 pg2
- McQs642Ta pg7
- Mcqs STA642 current solved2022 pg7
- By raza sta 642 ans with question pg12
- Sta642 26march2022 quiz Everblue pg5
- STA642 Current Final correct mcq by Hussnain Ali pg15

STA642 Mega For Final Term March 2022

Prepared by Everblue

Subjective (14 files)

- Sta642 108 to 185 handwritten notes merge in 1 file pg81
- STA642 short Notes written by Mahar.Afaq Safdar .Muhammadi pg13
- M@!!k-ST A642 distribution notes pg33
- STA642 Spring 2021 paper solution Mahar.Afaq Safdar .Muhammadi pg31
- STA642 Current Subjective 2022 mark by Hussnain Ali pg31
- STA642 Example By Mahar Afaq Safdar .Muhammadi pg40
- STA642 Important Final Term By Moin Akhtar pg54
- Stat 642 Important final Question by Sir Sikandar pg60
- STA642 mega file for final term by Haseeb Sahib pg170
- STA642 Mega For Final Term All in 1 File 2022 pg308
- Sta642 current file 99% paper solved pg183
- STA642 Current quiz+paper 14to25March2022 pg130
- Sta642 latest final papers Rashid Hussain pg44
- STA642 Current Subjective 2022 mark by Hussnain Ali pg31

STA642 Mega For Final Term March 2022

Prepared by Everblue

Lec (22-45), Topic (108-185)

Objective (11 files)

- Sta642 imP fOrMuLaS pg7
- Sta642 formula pg8
- Sta642 formulas pg10
- STA642-formulas pg4
- STA 642 random formulas pg10
- STA642 current paper mcqs 15march2022 pg2
- McQs642Ta pg7
- Mcqs STA642 current solved2022 pg7
- Byraza sta 642 ans with question pg12
- Sta642 26march2022 quiz Everblue pg5
- STA642 Current Final correct mcq by Hussnain Ali pg15

Imp Formulas: Sta642

ALL PDFs: (for continuous)

1- Continuous Uniform Distribution:-

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{for } a \leq x \leq b \\ 0 & \text{for } x < a \text{ or } x > b. \end{cases}$$

2- Exponential Distribution:-

$$f(x; \lambda) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0, \lambda > 0 \\ 0 & x < 0 \end{cases}$$

3- Gamma Distribution:-

$$f(x) = \begin{cases} \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} e^{-x/\beta} & 0 < x < \infty, \alpha > 0, \beta > 0 \\ 0 & \text{elsewhere} \end{cases}$$

4- Beta Distribution:-

$$f(x, \alpha, \beta) = \frac{1}{B(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1}, \quad 0 \leq x \leq 1$$

where:

$$B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$$

5- Normal distribution:-

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \quad \text{for } -\infty < x < \infty$$

6- Standard Normal Distribution:-

$$f_Z(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}, \quad -\infty < z < \infty$$

7- Bivariate Normal Distribution:

$$f(x_1, x_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} e^{-\frac{1}{2(1-\rho^2)} \left[\left(\frac{x_1-\mu_1}{\sigma_1} \right)^2 - 2\rho \left(\frac{x_1-\mu_1}{\sigma_1} \right) \left(\frac{x_2-\mu_2}{\sigma_2} \right) + \left(\frac{x_2-\mu_2}{\sigma_2} \right)^2 \right]}$$

⇒ All PMFs (for discrete).

1- Discrete uniform Distribution:

$$P(x) = \begin{cases} \frac{1}{n} & x=1, 2, 3, \dots, n \\ 0 & \text{elsewhere} \end{cases}$$

2- Binomial Distribution:

$$P(x) = \begin{cases} \binom{n}{x} p^x (1-p)^{n-x} & x=0, 1, 2, \dots, n \\ 0 & \text{elsewhere} \end{cases}$$

3- Negative binomial Distribution:

$$P(X=x) = \binom{x-1}{r-1} p^r (1-p)^{x-r}, \quad x=r, r+1, r+2, \dots$$

4- Geometric Distribution:

$$P(X=x) = \binom{x-1}{0} p^1 (1-p)^{x-1} = (1-p)^{x-1} p, \quad x=1, 2, 3, \dots$$

5- Multinomial Distribution:

$$P(x_1, x_2, \dots, x_k, p_1, p_2, \dots, p_k) = \begin{cases} \frac{n!}{x_1! x_2! \dots x_k!} p_1^{x_1} p_2^{x_2} \dots p_k^{x_k} & \text{otherwise} \\ 0 & \end{cases}$$

6- Hypergeometric Distribution:-

$$P(X=x) = \frac{\binom{k}{x} \binom{N-k}{n-x}}{\binom{N}{n}}, \quad x=0,1,2,\dots,k \text{ if } k \leq n \\ \text{ \& } x=0,1,2,\dots,n \text{ if } n \leq k$$

7- Poisson Distribution:-

$$P(x) = \begin{cases} \frac{e^{-\lambda} \lambda^x}{x!}, & x=0,1,2,\dots, \lambda > 0 \\ 0 & \text{elsewhere} \end{cases}$$

⇒ All CDFs:-

1- Discrete uniform Distribution:-

$$P(X=x) = \begin{cases} \frac{1}{b-a+1}, & x=a, a+1, \dots, b-1, b \\ 0 & \text{elsewhere} \end{cases}$$

2- Continuous uniform Distribution:-

$$F(x) = \begin{cases} 0 & \text{for } x < a \\ \frac{x-a}{b-a} & \text{for } a \leq x \leq b \\ 1 & \text{for } x > b \end{cases}$$

3- Exponential Distribution:-

$$F(x; \lambda) = \begin{cases} 1 - e^{-\lambda x} & x \geq 0, \lambda > 0 \\ 0 & x < 0 \end{cases}$$
$$F(x; \theta) = \begin{cases} 1 - e^{-x/\theta} & x \geq 0, \theta > 0 \\ 0 & x < 0 \end{cases}$$

4- Gamma Distribution:-

$$F(x) = \begin{cases} \frac{1}{\Gamma(\alpha)} \gamma\left(\alpha, \frac{x}{\beta}\right) & \alpha > 0, \beta > 0 \\ 0 & \text{elsewhere} \end{cases}$$

5- Beta Distribution.

$$F(x) = I_x(\alpha, \beta) = \frac{\int_0^x t^{\alpha-1} (1-t)^{\beta-1} dt}{B(\alpha, \beta)}, \quad 0 \leq x \leq 1, \alpha, \beta > 0.$$

6- Normal Distribution.

$$F(x) = \frac{1}{2} \left[1 + \operatorname{erf} \left(\frac{x-\mu}{\sigma\sqrt{2}} \right) \right] \text{ for } -\infty < x < \infty$$

erf defined as:

$$\operatorname{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z e^{-t^2} dt.$$

⇒ MEAN:

1) Discrete U.D:-

$$\mu = E(X) = \frac{n+1}{2}$$

2) Binomial Theorem:

$$\mu = np.$$

3) Negative Binomial Dist.

$$E(X) = \mu = \frac{r(1-p)}{p} = \frac{rq}{p}$$

4) Geometric Dist.

$$\mu = E(X) = \frac{1}{p}$$

5) Multinomial Dist.

$$\mu = E(X_i) = np_i$$

6) Hypergeometric Dist.

$$\mu = E(X) = \frac{nk}{N}$$

7) Poisson Dist.

$$\mu = E(X) = \lambda \quad (\text{same (mean and var)})$$

8) Continuous U.D.:

$$\mu = E(X) = \frac{a+b}{2}$$

9) Exponential Dist.

$$\mu = E(X) = \frac{1}{\lambda}$$

10) Gamma Dist.:

$$\mu_X = E(X) = \alpha\beta$$

11) Beta Distributions.

$$\mu = E(X) = \frac{\alpha}{\alpha + \beta}$$

12) Normal Dist.:

$$M(t) = E(e^{tx}) = e^{\mu t + \frac{1}{2}\sigma^2 t^2} \quad \text{for } -\infty < t < \infty$$

13) Standard Normal Dist.:

$$\text{mean} = \mu = E(X) = \int_{-\infty}^{\infty} x f_X(x) dx = 0$$

$\Rightarrow$ Variance.

1- Discrete U.D.:

$$\sigma^2 = E(X^2) - (E(X))^2 = \frac{n^2 - 1}{12}$$

2- Binomial Theorem:

$$\sigma^2 = np(1-p) = npq$$

3- Neg Binomial dist.:

$$\sigma^2 = \frac{r(1-p)}{p^2} = \frac{rq}{p^2}$$

4- Geometric Dist.

$$\sigma^2 = \frac{1-p}{p^2} = \frac{q}{p^2}$$

5- Multinomial Dist.

$$\sigma^2 = np_i(1-p_i) = np_i q_i$$

6- Hypergeometric Dist.

$$\sigma^2 = n \cdot \frac{k}{N} \cdot \frac{(N-k)}{N} \cdot \frac{(N-n)}{N-1}$$

7- Poisson Distr.

$$\text{Var} = \sigma^2 = \lambda \quad (\text{mean} = \text{Var}).$$

8- Continuous U.D.

$$\sigma^2 = \text{Var}(X) = \frac{(b-a)^2}{12}$$

9- Exponential Dist.

$$\text{Var}(X) = \sigma^2 = \frac{1}{\lambda^2}$$

10- Gamma Dist.

$$\text{Var}(X) = \sigma^2 = \alpha \beta^2$$

11- Beta Dist.

$$\sigma^2 = \frac{\alpha \beta}{(\alpha + \beta)^2 (\alpha + \beta + 1)}$$

12- Standard Normal Dist.

$$\text{Var}(Z) = \sigma^2 = 1$$

⇒ Mode

1- ~~Gamma~~ Gamma Distribution.

$$\text{Mode} = (\alpha - 1)\beta$$

2- Beta Dist:-

$$\text{Mode} = \frac{\alpha - 1}{\alpha + \beta - 2}$$

3- Normal Dist.

$$\text{Mode} = \mu, \text{ Median} = \mu$$

⇒ MGF

1- Discrete U.D:-

$$M(t) = E(e^{tx}) = \frac{e^t - e^{(n+1)t}}{n(1 - e^t)}$$

2- Binomial Dist.

$$M(t) = (1 - p + pe^t)^n$$

3- Neg Binomial Dist.

$$M(t) = \left(\frac{(1-p)}{1-pe^t} \right)^r \text{ for } t < -\log p$$

4- Multinomial Dist.

$$M_x(t) = \left(\sum_{i=1}^k p_i e^{t_i} \right)^n$$

5- Trinomial Dist.

$$M_x(t) = (p_1 e^{t_1} + p_2 e^{t_2} + p_3 e^{t_3})^n$$

6- Poisson Dist:-

$$M(t) = e^{m(e^t - 1)}$$

7- Exponential Dist:-

$$M(t) = E(e^{tx}) = \frac{\lambda}{\lambda - t}$$

8- Gamma Dist:-

$$M(t) = (1 - \beta t)^{-\alpha} \text{ for } t < \frac{1}{\beta}$$

9- Bivariate Normal Dist:-

$$M_x(t) = e^{ut_1 + ut_2 + \frac{1}{2}(\sigma_1^2 t_1^2 + 2\rho\sigma_1\sigma_2 t_1 t_2 + \sigma_2^2 t_2^2)}$$

$$-\infty < t_1 < \infty \\ ; -\infty < t_2 < \infty$$

~~10~~ $\Rightarrow$ Cumulant $\therefore$

1- Multinomial:-

$$K_X(t) = \log[M_X(t_1, t_2, \dots, t_k)] \\ = n \log\left(\sum_{i=1}^k p_i e^{t_i}\right)$$

2- Normal dist:-

$$K(t) = \ln E(e^{tx}) = \ln e^{ut + \frac{1}{2}\sigma^2 t^2}$$

$$K_1 = \mu, K_2 = \sigma^2$$

$$K_3 = K_4 = \dots = 0$$

TAUQEER
0303-0254654

STA - 642

PMF = Probability Mass function

PDF = Probability Density function

CDF = Cumulative distribution function

MGF = Moment generating function

CGF = Cumulant generating function

* Parameters $\Rightarrow$

Discrete uniform distribution (a, b)

Binomial distribution (p, n)

Gamma distribution (α, β)

Beta distribution (α, β)

Normal distribution (μ, σ)

①

Discrete uniform distribution $\Rightarrow$

PMF
$$P(X) = \begin{cases} \frac{1}{n} & x = 1, 2, \dots, n \\ 0 & \text{elsewhere} \end{cases}$$

Mean
$$\mu = \frac{n+1}{2}$$

Variance
$$\text{Var}(X) = \frac{n^2 - 1}{12}$$

MGF
$$M(t) = \frac{e^t - e^{-(n+1)t}}{n(1 - e^t)}$$

②

Binomial distribution $\Rightarrow$

PMF $P(X) = \begin{cases} \binom{n}{x} p^x (1-p)^{n-x} & x=0,1,2,\dots,n \\ 0 & \text{else where} \end{cases}$

Mean $\mu = np$

Variance $\text{Var}(X) = np(1-p)$

MGF $M(t) = [(1-p) + pe^t]^n$

③

Negative binomial distribution $\Rightarrow$

PMF $P(X=x) = \binom{x-1}{r-1} p^r (1-p)^{x-r}$

mean $\mu = \frac{r(1-p)}{p}$

Variance $\sigma^2 = \frac{r(1-p)}{p^2}$

MGF $M(t) = \left(\frac{p}{1-(1-p)e^t} \right)^r$

Parameter = r and p

④

Geometric Distribution $\Rightarrow$

PMF $P_X(x) = (1-p)^{x-1} p$

Mean $\mu = \frac{1}{p}$

Variance $\sigma^2 = \frac{1-p}{p^2}$

Parameter $\Rightarrow$ p only

⑤

Multinomial distribution

Mean $E(x_i) = np_i$

Variance $Var(x_i) = np_i(1-p_i)$

⑥

Hypergeometric distribution

PMF $P(X=x) = \frac{\binom{K}{x} \binom{N-K}{n-x}}{\binom{N}{n}}$

Mean $E(x) = n \frac{K}{N}$

$E(x) = np$

$\therefore p = \frac{K}{N}$

Variance $Var(x) = n \frac{K}{N} \frac{N-K}{N} \frac{N-n}{N-1}$

Parameters x, N, n, K

⑦

Poisson distribution

PMF $P(x) = \begin{cases} \frac{e^{-\lambda} \lambda^x}{x!} \\ 0 \end{cases}$

Mean $\mu = \lambda$

Variance $Var(x) = \lambda$

MGF $M(t) = e^{\lambda(e^t - 1)}$

Parameter λ only

⑧

Continuous Uniform / Rectangular distribution

Mean $E(x) = \frac{a+b}{2}$

Variance $Var(x) = \frac{(b-a)^2}{12}$

⑨

Exponential distribution

Mean $\mu = \frac{1}{\lambda}$

Variance $Var(x) = \frac{1}{\lambda^2}$

Covariance $Cov = 1$

⑩

Gamma distribution

Mean $\mu_x = \alpha \beta$

Variance $\sigma^2_x = \alpha \beta^2$

⑪

Beta distribution

Mean $\mu = \frac{\alpha}{\alpha + \beta}$

Variance $Var(x) = \frac{\alpha \beta}{(\alpha + \beta)^2 (\alpha + \beta + 1)}$

⑫

Normal distribution

Mean $= \mu$

Variance $= \sigma^2$

$\mu = \frac{\sum x}{n}$

+

1) Sum for natural numbers (first n)

$$\frac{n(n+1)}{2}$$

2) Sum for finite geometric series.

$$\text{Sum} = \frac{a(1-r^n)}{1-r}$$

3) Sum for Infinite geometric series

$$\text{Sum} = \frac{a}{1-r}$$

4) Variance:-

$$\text{Var } X = [E(X)^2] - [E(X)]^2$$

also $E\left[\left(\frac{X-\mu}{\sigma}\right)^2\right] = 1$ OR $\text{Var}\left[\frac{X-\mu}{\sigma}\right] = 1$

5) Standard deviation

$$E\left(\frac{X-\mu}{\sigma}\right) = 0$$

6) Measure of Skewness

$$= \frac{\mu_3}{\sigma^3} \quad \text{first moment ratio} \quad \beta_1 = \frac{\mu_3^2}{\mu_2^3}$$

7) Measure of Kurtosis

$$\beta_2 = \frac{\mu_4}{\mu_2^2} \quad \text{Second moment ratio}$$

Statistics

Probability = $\frac{\text{number of events}}{\text{number of sample space}}$

$$P = \frac{n(E)}{n(S)}$$

when set is finite, referring to discrete random variable.

when set is infinite, it is Continuous Random Variable

Formula for Infinite Geometric Series.

$$\text{Sum. G.S} = \frac{a}{1-r}$$

Prob. Mass function (pmf) / Discrete case.

$$P_x(x) = P[X=x] \quad x \in D$$

$$1) \quad 0 \leq P_x(x) \leq 1$$

$$2) \quad \sum_{x \in D} P_x(x) = 1$$

Discrete

CDF (cumulative distribution function)

CDF or $F(x)$

if pmf is $\frac{1}{6}$,

$$\text{CDF / } F(x) = F(6) = P(X \leq 6) = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6}$$

$$\text{Pos. } p = \frac{m}{n} = \frac{\text{obtained outcomes}}{\text{total number of outcomes}}$$

Sum of first n -natural numbers.

$$\text{Sum} = \frac{n(n+1)}{2}$$

⊗ CDF $F(x) \rightarrow$ For Continuous.

$$F_x(x) = \int_{-\infty}^x f_x(t) dt$$

PDF (Prob. density function) / $f(x)$

In case of Continuous

$$\frac{d}{dx} F_x(x) = f_x(x)$$

Prop. of PDF (Continuous case).

$$\int_{-\infty}^{\infty} f_x(x) dx = 1 \quad \& \quad f_x(x) \geq 0$$

$$\text{Also } P(a < x < b) = \int_a^b f_x(t) dt = F_x(b) - F_x(a)$$



Probability Generating function
In Discrete case.

$$G(z) = E(z^X) = \sum_{x=0}^{\infty} P(x) z^x$$

Probability for Poisson distribution

$$p(x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

probability mass function of any
Random Variable X .

$$P(K) = P(X=K) = \frac{G^{(K)}(0)}{K!}$$

take derivative of Prob. generating function

as $G^{(K)} \Rightarrow$ put $X=0$ in that function

& divide random variable's factorial

Expectation of Random Variable X

$$E(X) = G'(1^-)$$

$$G'(1^-) = \lim_{z \rightarrow 1} G'(z) \text{ from below}$$

$\Rightarrow$ Mean of Poisson is equal to λ .
also = to its expectation value.

$\Rightarrow$ Variance of Random Variable.
 $\text{Var}(X)$

$$= G''(1^-) + G'(1^-) - [G'(1^-)]^2$$

PGFs. formulas.

i) Constant r.v.

$$G_X(z) = E(z^X) = z^c$$

$$\text{if } P_c = 1, P_k = 0, K \neq c$$

ii) Bernoulli r.v.

$$\text{if } P_1 = p, P_0 = q, P_k = 0, K \neq 0, 1$$

$$G_X(z) = E(z^X) = q + pZ$$

iii) Geometric r.v.

$$\text{if } P_k = pq^{k-1}, K=1, 2, \dots; q=1-p$$

$$G_X(z) = \frac{pZ}{1-qZ} \quad \text{if } |z| < q^{-1}$$

iv) Binomial s.v.

$$X = \text{Bin}(n, p)$$

$$G_X(z) = (q + pz)^n$$

v) Poisson r.v.

$$X = \text{Poisson } \lambda$$

$$G_X(z) = \sum_{k=0}^{\infty} \frac{1}{k!} \lambda^k e^{-\lambda} z^k = e^{\lambda(z-1)}$$

vi) Negative binomial

$$X = \text{Neg Bin}(n, p)$$

$$G_X(z) = \sum_{k=0}^{\infty} \binom{k-1}{n-1} p^n q^{k-n} z^k$$

$$= \left(\frac{pz}{1-qz} \right)^n \quad \text{if } |z| < q^{-1}$$

Expectation = Mean of distribution (linear)

$$E(X) = \sum X \text{ Pdf}$$

Pdf of Linear Combination

$$\int_{-\infty}^{\infty} w(x) dx = 1$$

$$\text{pdf } w(x) = w(x) = C_1 f_1(x) + C_2 f_2(x) + \dots + C_n f_n(x)$$

$$E(X) = \sum X \text{ Pdf} = \mu$$

$$\mu = \sum_{i=1}^n C_i \mu_i$$

$\Rightarrow$ Expectation Value of R.v

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx$$

⇒ Mode.

value of x that max. pdf

⇒ Median. (discrete & continuous)

if $P(X \leq x) \leq \frac{1}{2}$ & $P(X \leq x) \geq \frac{1}{2}$

Then, $\exists$ only one such x , which is called the Median of distribution.

In Continuous case.

Median is obtained by Integrating pdf from $-\infty$ to M & equating it to $\frac{1}{2}$

⇒ Expectation of R.V for discrete.

$$\sum |x| P(x) < \infty$$

$$E(X) = \sum_x x P(x)$$

⇒ Expectation of R.V for Continuous case

~~$\sum_{-\infty}^{\infty} x f(x) dx$~~ $\int_{-\infty}^{\infty} x f(x) dx$

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx$$

$$E(X) = \mu \quad \Leftrightarrow \quad \sum x p(x) = \mu$$

⇒ Expectation of function of R.V.
(with pdf)

Continuous case: - let $y = g(x)$

$$\int_{-\infty}^{\infty} |g(x)| f(x) dx < \infty$$

$$E(Y) = \int_{-\infty}^{\infty} g(x) f(x) dx$$

Expectation of function of S.V

Discrete case:- with pmf

$$S_x = \sum_{x \in S_x} |g(x)| p_x(x) < \infty$$

$$E(Y) = \sum_{x \in S_n} g(x) p_x(x)$$

Mth Moments.

$$\mu'_m = E(X-a)^m \text{ (about Arbitrary origin)}$$

For Continuous: $\int_{-\infty}^{\infty} (x-a)^m f(x) dx$

For discrete: $\sum (x-a)^m p(x) dx$



mth moment about Mean ~~μ~~

$$\mu_m = E(X - \mu)^m \text{ is central moment}$$

Cont:-

$$\int_{-\infty}^{\infty} (x - \mu)^m f(x) dx$$

discrete:

$$\sum (x - \mu)^m p(x) dx$$



mth Moment about the origin.

$$\mu'_m = E(X^m)$$

Continuous:

$$\int_{-\infty}^{\infty} x^m f(x) dx$$

discrete:-

$$\sum x^m p(x) dx$$

Moment generating function:

$$M(t) = E(e^{tx}) \quad -h < t < h$$

Algebraic Expression of some MGFs
 → for discrete

Binomial $q + pe^t$

Geometric $\frac{pe^t}{1 - ve^t}$

Binomial $(q + pe^t)^n$

Poisson λ $e^{\lambda(e^t - 1)}$

-ve. binomial $\frac{(q)^x}{(1 - pe^t)^x}$

Uniform (discrete) $\frac{e^{at} - b^{(b+1)t}}{(b-a+1)(1 - e^t)}$

Continuous distribution.

Uniform $\frac{e^{tb} - e^{ta}}{t(b-a)}$

Normal $e^{t\mu + \frac{1}{2}\sigma^2 t^2}$

Chi-Square $(1-2t)^{-k/2}$

Gamma $(1-t\theta)^{-k} \quad \forall t < \frac{1}{\theta}$

Exponential $(1-t\lambda)^{-1} \quad (t < \lambda)$

MGF.

Laplace

$$\frac{e^{tu}}{1-b^2 t^2}$$

Cauchy

don't exist.

Moment generating function

$$M(t) = E(e^{tx})$$

Characteristic function

$$\phi_x(t) = E(e^{itx})$$

Continuous case

$$\phi_x(t) = E(e^{itx}) = \int_{-\infty}^{\infty} e^{itx} f(x) dx$$

Characteristic functions

Distribution

Cumulative function $\Phi(t)$

Bernoulli

$$(q + pe^t)$$

$$\{q = 1-p\}$$

Binomial

$$(q + pe^t)^n$$

Poisson

$$e^{\lambda(e^t - 1)}$$

Geometric

$$\frac{pe^{it}}{1 - qe^{it}}$$

Uniform (discrete)

$$\frac{e^{a+it} - e^{(b+1)+it}}{(b-a+1)(1 - e^{it})}$$

Exponential

$$(1 - it\lambda^{-1})^{-1}$$

Normal N.

$$e^{itu} - \frac{1}{2} \sigma^2 t^2$$

⇒ Joint probability mass function $0 < P_{X_1, X_2} < 1$

$$P_{X_1, X_2}(x_1, x_2) = P[X_1 = x_1, X_2 = x_2] \quad \sum \sum P_{X_1, X_2} = 1$$

$$\text{OR } P[(X_1, X_2) \in B] = \sum \sum_B P_{X_1, X_2}(x_1, x_2)$$

⇒ Joint Cumulative distribution function

$$F_{X_1, X_2}(x_1, x_2) = P[\{X_1 \leq x_1\} \cap \{X_2 \leq x_2\}]$$

$$\text{OR } P[X_1 \leq x_1, X_2 \leq x_2]$$

⇒ Random vector (Discrete)

$$X_1(\omega) = x_1, \quad X_2(\omega) = x_2$$

⇒ Random vector Continuous case

$(x_1, x_2) \in D$, D is continuous type if its cdf $F_{X_1, X_2}(x_1, x_2)$ is continuous

$$F_{X_1, X_2}(x_1, x_2) = \int_{-\infty}^{x_1} \int_{-\infty}^{x_2} f_{X_1, X_2}(w_1, w_2) dw_1 dw_2$$

⇒ Joint Probability density function

Properties -

$$1) f_{X_1, X_2}(x_1, x_2) \geq 0$$

$$2) \iint_D f_{X_1, X_2}(x_1, x_2) dx_1 dx_2 = 1$$

1. Geometric distribution is a special case of which distribution

Ans negative binomial distribution

2. Instead of counting the number of successes we are interested to count the numbers of trials until a success is obtained. Which probability distribution should be used in this case

Binomial distribution

3. Choose the correct formula of mean of multinomial distribution

$E(X_i) = np_i$

4. The domain $0 < x < y < 1$ can be Re-written as

$0 < x < 1, x < y < 1$

5. X and Y are independent if and only if

$P = 0$

6. In normal distribution (sigma square) denotes the

Variance

7. For the standard normal distribution $E(z)$ is equal to which value

$E(Z) = 0$

8. Choose the correct number of parameters(s) for normal distribution

μ, σ

9. What is relationship b/w the mean, median and mode of a normal distribution

Are equal

10. In normal distribution, the all odd order moments about mean are

Zero

11. The first Raw moment of the normal distribution is equal to

12. The first Raw moment of the normal distribution is equal to

10. In normal distribution, the all odd order moments about mean are

11. The first Raw moment of the normal distribution is equal to

Expected value or mean

1 2. The first cumulant $k'_1(O)$ of normal distribution is

$K_1 = \mu$

1 3. All the higher cumulant of a normal distribution is

zero

1 4. The bivariate normal distribution has

Two independent variables

1 5. The bivariate distribution can be expressed as

$P(x,y)$

1 6. Any subset of a sample space is called

Event

1 7. The quantile function is called

Percentile function or inverse of cumulative distribution

1 8. If the distribution satisfies the definition of "random variable" even though it does not appear random in the every day sense of the word; it is considered as

Degenerate

1 9. The expected value of e^{itx} is called

Characteristic function

20. Joint probability distribution of two variables is called

Joint probability distribution

21. In one-to-one transformation, for two old variables, we get

Two linearly independent variables

22. In which distribution the successive draws are conducted through sampling without replacement

Hypergeometric distribution

23. The variance of the hypergeometric distribution is given by

$$\frac{(K)(N-K)(X)(n-x)/(N)}{(n)}$$
24. The mean and variance in poisson distribution is given by

Mean and variance are equal

-
25. Which distribution has a unique property that its mean and variance are equal

Poisson

26. The formula

$$a + b/2$$

— is used to find the mean of which distribution

Continuous uniform distribution

27. For the exponential distribution, λ : A then 0?
 if

Given same value and find theta = lambda

27. The given pdf belongs to which distribution

(Gamma distribution ka pdf tha) 29.

For any fixed value of the scale parameter β as α increase, the shape of the
distribution tends to

30. The limit of probability density function of the β -distribution are:

0 less than or equal to x less than or equal to 1

31. The shape parameters of beta distribution is
Alfa and beta

32. The 2-parameters of uniform distribution are
a and b

33. Which two parameters determine the shape of binomial distribution

P and n

34. What is the formula of finding the variance of binomial distribution

Sigma power alfa = npq

35. For the mth moment about the origin if $m=1$ then we obtain

Mean

36. For the mth moment about the origin if $m=2$ then we obtain

Variance

37 Consider the equation of circle $(x_1 - c_1)^2 + (x_2 - c_2)^2 = a^2$ which is
showing the value of radius

a

38. How many parameters are in binomial normal distribution

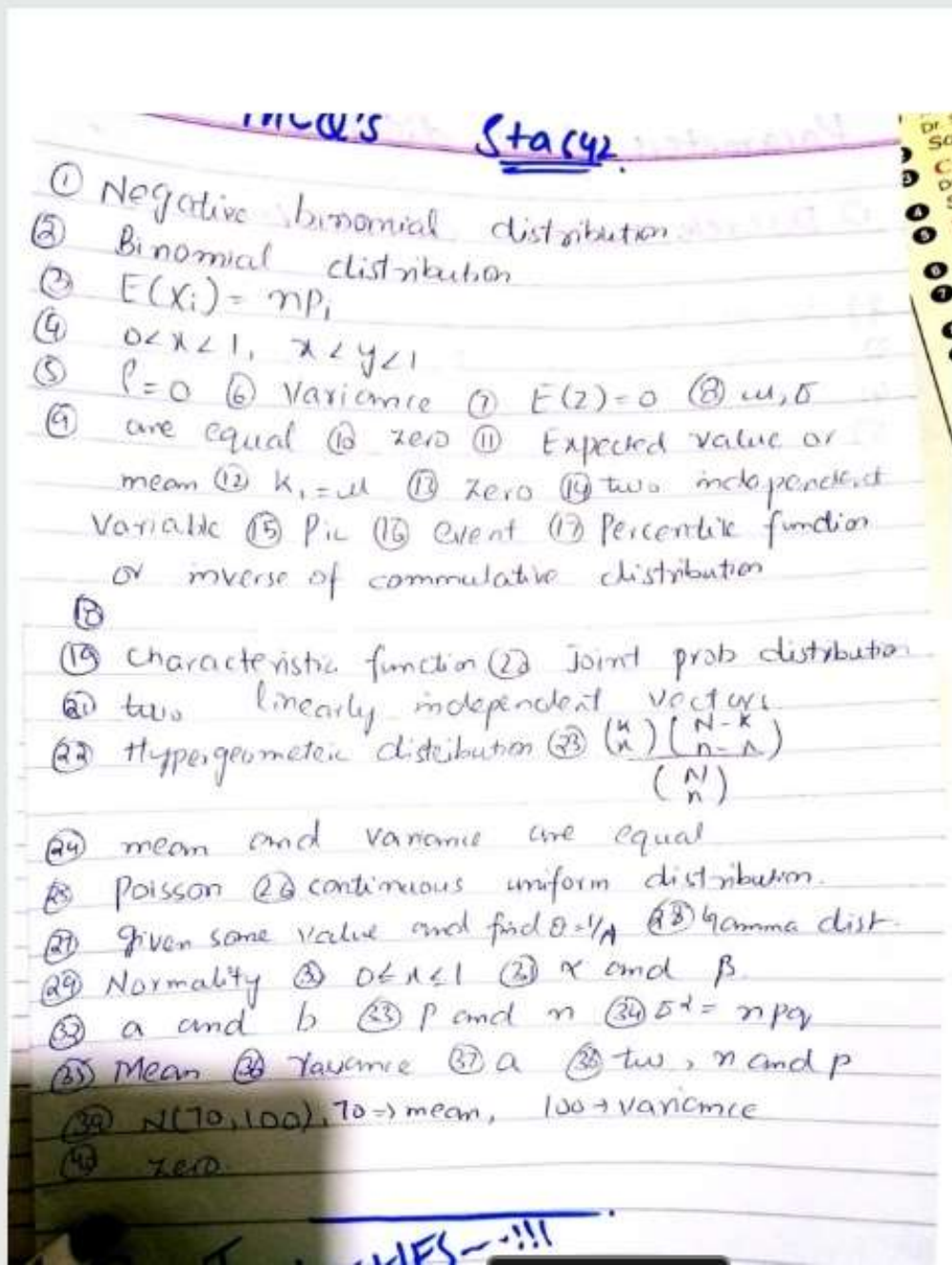
39. A variable is normally distributed with

NC70, 100), what is the mean of this distribution

70 is the mean and 100 is the variance

40. In the case of a Bivariate Normal distribution the random variables X and Y are independent if and only if the correlation coefficient

Zero



Parameters of all distributions:-

1) Discrete uniform distribution:-

a and b.

2) Binomial $\Rightarrow$ p and n.

3) Gamma $\Rightarrow$ α and β .

4) Beta $\Rightarrow$ α and β

5) Normal dist $\Rightarrow$ $N(\mu, \sigma)$

_____.

STA642

Current paper

15th-march-22

Moqs

1. Geometric distribution is a special case of which distribution
2. Instead of counting the number of successes we are interested to count the numbers of trials until a success is obtained. Which probability distribution should be used in this case
3. Choose the correct formula of mean of multinomial distribution
4. The domain $0 < x < y < 1$ can be re-written as
5. X and Y are independent if and only if
6. In normal distribution (sigma square) denotes the
7. For the standard normal distribution $E(z)$ is equal to which value
8. Choose the correct number of parameters(s) for normal distribution
9. What is relationship b/w the mean, median and mode of a normal distribution
10. In normal distribution, the all odd order moments about mean are
11. The first raw moment of the normal distribution is equal to
12. The first cumulant $k'(0)$ of normal distribution is
13. All the higher cumulant of a normal distribution is
14. The bivariate normal distribution has
15. The bivariate distribution can be expressed as
16. Any subset of a sample space is called
17. The quantile function is called
18. If the distribution satisfies the definition of "random variable" even though it does not appear random in the every day sense of the word; it is considered as
19. The expected value of e^{itx} is called
20. Joint probability distribution of two variables is called
21. In one-to-one transformation, for two old variables, we get
22. In which distribution the successive draws are conducted through sampling without replacement
23. The variance of the hypergeometric distribution is given by
24. The mean and variance in poisson distribution is given by

25. Which distribution has a unique property that its mean and variance are equal
26. The formula $\frac{a+b}{2}$ is used to find the mean of which distribution
27. For the exponential distribution, if $\theta = 1/\lambda$ then θ ?
28. The given pdf belongs to which distribution (Gamma distribution ka pdf tha)
29. For any fixed value of the scale parameter β as α increase, the shape of the distribution tends to
30. The limit of probability density function of the β -distribution are
31. The shape parameters of beta distribution is
32. The 2-parameters of uniform distribution are
33. Which two parameters determine the shape of binomial distribution
34. What is the formula of finding the variance of binomial distribution
35. For the m th moment about the origin if $m=1$ then we obtain
36. For the m th moment about the origin if $m=2$ then we obtain
37. Consider the equation of circle $(x_1 - c_1)^2 + (x_2 - c_2)^2 = a^2$ which is showing the value of radius
38. How many parameters are in binomial normal distribution
39. A variable is normally distributed with $N(70, 100)$, what is the mean of this distribution
40. In the case of a Bivariate Normal distribution the random variables X and Y are independent if and only if the correlation coefficient ρ is equal to

Paper 15 March 2022 Subject

1. Negative Binomial

2.

3. multinomial

$$\frac{n!}{x!(n-x)!} p_1^{x_1} (1-p)^{n-x_1}$$

$$\frac{n!}{x_1! x_2!} p_1^{x_1} p_2^{x_2}$$

4. Domain $0 \leq x < 1$ $0 < y < 1$

5. X, Y independent

$$F(x, y) = F_x(x) \cdot F_y(y)$$

6. σ^2 Variance $\mu = \text{Mean}$

$\sigma = \text{STD}$

7. Standard Normal Distribution

$$E(Z) = \int_{-\infty}^{\infty} Z f_Z(Z) dZ$$

8. Normal Dis $f(n) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$
Mean, Std.

10) odd moment $= 0$

⑪ Raw moment $m''(0) \downarrow$
 $m'(0) = E(X) = \mu$ $E(X^2) = \sigma^2 + \mu^2$

Variance $= \sigma^2$

⑫ Cumulant $k(0) = \mu$
 $k''(0) = \sigma^2$
 $k'''(0) = 0$

⑬ Higher order cumulant $= 0$

⑭

⑮

⑯ Subset $\longrightarrow$ Space $\longrightarrow$ Event

⑰ Quantile function
inverse cdf

⑱ A degenerate

$$19) E(e^{itx}) = \int_{-\infty}^{\infty} e^{itx} f(x) dx \quad (98)$$

< Characteristic function >

20) Joint Probability of Two
< conditional >

21) Two new

22) without replacement $\rightarrow$ Hyper Geo
 $\downarrow$

23) Variance $n \cdot \frac{K}{N} \cdot \frac{N-K}{N} \cdot \frac{N-n}{N-1}$ or NPq

24) Mean & variance of Poisson = λ
 $\downarrow$

25) Equal $\rightarrow$ Poisson

$$26) \frac{a+b}{2}$$

$$27) \theta = \frac{1}{\lambda} \Rightarrow \lambda = \frac{1}{\theta}$$

28) Gamma pdf $f(x) = \frac{1}{\beta^2} x e^{-x/\beta}$

29) tends to normality (158)

30) Limits $0 \leq x < 1 : \alpha, \beta > 2 \leq n$ (162)

31) α and β are shape parameter

32) Uniform distribution parameter
 a and b

33) Shape of binomial : p and n (

34) Formula for variance of Binom.
 $\mu = np$ $\sigma^2 = np(1-p)$

35) m^{th} moment about origin
Topic $\mu'_m = E(X^m)$ $\mu'_1 = E(X)$

36 $n=2$ $\mu_2 = E(X-\mu)^2$

37. $(x_1 - c_1)^2 + (x_2 - c_2)^2 = a^2$
circle (c_1, c_2) with radius

38. Binomial normal n, p

39. $N(70, 100)$

40) $\rho = 0$ independent

- 1) Any subset of a sample is called event
- 2) Generally, Probability density function is symbolically denoted by $f(x)$
- 3) Discrete Probability distribution in which outcome is very small with a very small period of time is ~~classified~~ classified as

a) Poisson distribution b) Posterior distribution
c) Cumulative distribution d) Normal distribution

- 4) The domain $0 < x < y < 1$ can be re-written as

a) $0 < x + y < 1$ b) $0 < y < 1, y < x < 1$
c) $0 < x < 1, y < x < 1$ d) $0 < x < 1, x < y < 1$

- 5) $K(t)$ is equal to a) $E(M(t))$ b) $Var(M(t))$

c) $\log M(t)$ d) $M(t)$ (Topic 6.5)

- 6) The normal curve is asymptotic to which ~~axis~~ axes?

a) X-axis b) Y-axis c) Z-axis d) Both X-axis and Y-axis

- 7) Which two parameters determine the shape of binomial distribution?

a) n and p b) n and x c) p and q d) p and x

- 8) For a data we find correlation coefficient,

Now we add ^{the} digit 5 in all data values and then again computed the correlation, The new correlation coefficient

a) will decrease b) will increase by 5%

c) will remain same d) will increase

- 9) If random variable X is infinite then its expectation does not exist (Topic 89)
- 10) Choose the correct parameters of Poisson distribution λ (Topic 141)
- 11) In which distribution, the numbers of trial are NOT fixed in advance.
 a) Binomial distribution b) Bernoulli distribution
 c) Hypergeometric distribution d) Geometric distribution
- 12) Choose the correct numbers of normal distribution
 a) 1 b) 2 c) 3 d) 4
- 13) For the joint PMF $P_{X,Y}(x_1, x_2)$ we may obtain the Marginal PMF of Y_1 by Summing on Y_2 (Topic 96)
- 14) For the standard normal distribution, $F(z)$ is equal to which value? a) 0 b) 1 c) -1 d) 100 (Topic 166)
- 15) The limit of Probability density function of the β distribution are $0 \leq x \leq 1$ and $\alpha, \beta > 0$ (Topic 164)
- 16) In one-to-one transformation, for two old variables, we get d)
 a) one new variable b) three new variable
 c) four new variable d) two new variable
- 17) The Shape parameters of beta distribution are a) α and β b) α and $-\beta$
 c) $1-\alpha$ and $-\beta$ d) α and β (Topic 162)
- 18) The formula $\frac{a+b}{2}$ is used to find the MEAN of which distribution Continuous Uniform Distribution (Topic 151)

19) The given Pdf belongs to which distribution
$$f(x) = \begin{cases} \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} e^{-x/\beta} & 0 \leq x < \infty, \alpha > 0, \beta > 0 \\ 0 & \text{elsewhere} \end{cases}$$

Gamma distribution (Topic 158)

19) In which distribution, the successive draws are conducted through "Sampling without replacement".

Hypergeometric distribution (Topic 137)

20) What is the formula of finding the MEAN of binomial distribution np (Topic 124)

21) For any fixed value of the scale parameter β , as x increases, the shape of distribution tends to Normality shape (Topic 158)

22) X and Y are independent if and only if

$$F_{XY}(x, y) = F_X(x) F_Y(y) \quad (\text{Topic 185})$$

23) Two parameter of Uniform distribution are

a and b (Topic 119, 151)

24) What is the correct formula of mean of the

Hypergeometric distribution? $\frac{N \cdot K}{N}$ (Topic 139)

25) The experiment continues until r successes are observed, where r is specified in advance.

This is an example of which distribution

Negative Binomial distribution (Topic 129)

26) What is relationship between the mean, median and mode of a normal distribution.

Always same

(Topic 171)

- 27) How many parameters are in the Bivariate Normal Distribution five (Topic 186)
- 28) The bivariate Normal distribution can be expressed as $N(\mu_x, \mu_y, \sigma_x^2, \sigma_y^2, \rho)$ (Topic 186)
- 29) Consider the equation of Circle $(x_1 - c_1)^2 + (x_2 - c_2)^2 = r^2$ (Topic 182)
Which of it's showing the value of radius? r
- 30) The Bivariate Normal distribution has Bell Shape (Topic 187)
- 31) In normal distribution, the all odd order moments about mean are Zero (Topic 174)
- 32) In general, the expected value of a Product is to the Product of the expected values, that is $E(XY) = E(X)E(Y)$ Not equal (Topic 47)
- 33) Which distribution possess the quality that its mean and standard deviation are equal Exponential distribution (Topic 154)
- 34) The first raw moment of the Normal distribution is equal to Mean $E(X)$ (Topic 176)
- 35) All higher ~~order~~th cumulants of a ~~binomial~~ normal distribution are equal to Zero (Topic 177)
- 36) In Bivariate Normal distribution, What is the ~~shape~~^{range} of limit of variable x and y - $-\infty < x, y < \infty$ (Topic 186)
- 37) The first ~~cumulant~~^(μ) $\kappa'(0)$ of normal distribution is Mean (Topic 65)

38) In the case of a Bivariate Normal Distribution, the random variable X and Y are independent if and only if the correlation coefficient (ρ) is equal to $\rho = 0$ (Topic 185)

39) Choose the correct formula of VARIANCE of Multinomial distribution: $n p_i (1 - p_i)$ (Topic 133)

40) The limits of probability density function of the β distribution are $0 \leq x \leq 1$, $\alpha, \beta > 0$ (Topic 162)

OBJECTIVE

MEGA FILE

FOR FINAL TERM (FALL 2021,22)

Course Code: STA642

Written BY:

SYED HUSSNAIN ALI NAQVI

MSC MATHEMATICS

Student 1 (15 March 2022)

Page # 02

Student 2 (26 March 2022)

Page # 04

BINOMIAL AND HYPERGEOMETRIC DISTRIBUTIONS

Page # 06

Current Solved ScreenShot (March 2022)

Page # 11

Student 1 (15 March 2022)

#	Statement	Answer
1	Geometric distribution is a special case of which distribution	Negative binomial distribution
2	Instead of counting the number of successes we are interested to count the numbers of trails until a success is obtained. Which probability distribution should be used in this case	Binomial distribution
3	Choose the correct formula of mean of multinomial distribution	$E(X_i) = nP_i$
4	The domain $0 < x < y < 1$ can be Re-written as	$0 < x < 1, x < y < 1$
5	X and Y are independent if and only if	$P = 0$
6	In normal distribution (sigma square) denotes the	Variance
7	For the standard normal distribution E(z) is equal to which value	$E(z)=0$
8	Choose the correct number of parameters(s) for normal distribution	μ, σ
9	What is relationship b/w the mean, median and mode of a normal distribution	Are equal
10	In normal distribution, the all odd order moments about mean are	Zero
11	The first Raw moment of the normal distribution is equal to	Expected value or mean
12	The first cumulant $k'(0)$ of normal distribution is	$K_1 = \mu$
13	All the higher cumulant of a normal distribution is	Zero
14	The bivariate normal distribution has	Two independent variable
15	The bivariate distribution can be expressed as	P_{ij}
16	Any subset of a sample space is called	Event
17	The quantile function is called	Percentile function or inverse of cumulative distribution
18	If the distribution satisfies the definition of “random variable” even though it does not Appear random in the everyday sense of the word; it is considered as	Degenerate
19	The expected value of e^{itx} is called	Characteristic function
20	Joint probability distribution of two variable is called	Join problem distribution
21	In-one-to-one transformation, for two old variables, we get	Two linear independent vectors
22	In Which distribution the successive draws are conducted through sampling without replacement	Hyper geometric distribution
23	The variance of the hyper geometric distribution is given by	$\frac{\binom{k}{x} \binom{N-k}{n-x}}{\binom{N}{n}}$
24	The mean and variance in poisson distribution is given by	Mean and variance are equal

25	Which distribution has a unique property that its mean and variance are equal	Poisson
26	The formula $\frac{a+b}{2}$ is used to find the mean of which distribution	Continues uniform distribution
27	For the exponential distribution , if $\theta=1/\lambda$ then θ ?	Give some value and find $\theta=1/\lambda$
28	The given pdf belongs to which distribution (Gamma distribution ka pdf tha)	Gamma distribution
29	For any fixed value of the scale parameter β as α increase, the shape of the distribution tends to	Normality shape
30	The limit of probability density function of the β -distribution are	$0 \leq x \leq 1$
31	The shape parameters of beta distribution is	α and β
32	The 2-parameters of uniform distribution are	a and b
33	Which two parameters determine the shape of binomial distribution	p and n
34	What is the formula of finding the variance of binomial distribution	$\sigma^2 = npq$
35	For the m^{th} moment about the origin if $m=1$ then we obtain	Mean
36	For the m^{th} moment about the origin if $m=2$ then we obtain	Variance
37	Consider the equation of circle $(x_1-c_1)^2+(x_2-c_2)^2=a^2$ which is showing the value of radius	a
38	How many parameters are in binomial normal distribution	μ, σ, n and p
39	A variable is normally distributed with $N(70,100)$, what is the mean of this distribution	$N(70,100)$ $70 \Rightarrow \text{mean}, 100 \Rightarrow \text{variance}$
40	In the case of a Bivariate Normal distribution the random variables X and Y are independent If and only if the correlation coefficient ρ is equal to	zero

Student 2 (26 March 2022)

#	Statement	A	B	C	D
1	In one-to-one transformation, for two old variables, we get a	One new variable	Three new variable	Four new variable	Two new variable
2	The shape parameters of beta distribution is	$-\alpha$ and β	α and $-\beta$	$-\alpha$ and $-\beta$	α and β
3	Discrete probability distribution in which outcome is very small period is classified as a	Poisson distribution	Posterior distribution	Cumulative distribution	Normal distribution
4	The domain $0 < x < y < 1$ can be Re-written as	$0 < x < 1$, $x < y < 1$	$0 < x+y < 1$	$0 < y < 1$ $y < x < 1$	$0 < x < 1$ $y < x < 1$
5	$K(t)$ is equal to	$E(M(t))$	$\text{Var}(M(t))$	$\text{Log } M(t)$	$M(t)$
6	The normal curve is asymptotic to which axis	x-axis	y-axis	z-axis	Both x-axis and y-axis
7	Which two parameter determine the shape of binomial distribution	p and n	p and x	p and q	q and x
8	For a data we find correlation coefficient, now we add a digit 5 in all the data values and then again computed the correlation, then new correlation coefficient is	Will decreases	Will increases by 5%	Will remain same	Will increase.
9	In which distribution, the numbers of trial are not fixed in advance.	Binomial distribution	Bernoulli distribution	Hyper geometric distribution	Geometric distribution
10	Chose the correct number of normal distribution	1	2	3	4
11	For the standard normal distribution $E(z)$ is equal to which value,	1	0	-1	100
12	The limit of probability density function of the β -distribution are	$0 \leq x \leq 1$ $\alpha, \beta > 0$			
13	For the join PMF $P_{Y_1 Y_2}(y_1 y_2)$ we say obtain the marginal PMF of y_1 by summing on	Y_2			
14	Any subset of a sample space is called	Event			
15	Generally probability density function is symbolically denoted by	$f_x(x)$			
16	If random variable X is infinite then its expectation	Does not exist			
17	Chose the correct parameters of poisson distribution	λ			
18	The formula $\frac{a+b}{2}$ is used to find the mean of which	Continues uniform distribution			

	distribution	
19	The given PDF belongs to which distribution $f(x) = \begin{cases} \frac{1}{\alpha\beta^\alpha} x^{\alpha-1} e^{-\frac{x}{\beta}} & 0 < x < \infty, \alpha > 0, \beta > 0 \\ 0 & \text{elsewhere} \end{cases}$	Gamma distribution
20	In which distribution the successive draws are conducted through sampling without replacement	Hyper geometric distribution
21	What is formula of finding the MEAN of binomial distribution	np
22	For any fixed value of the scale parameter β as α increase, the shape of the distribution tends to	Normality shape
23	X and Y are independent if and only if	$F_{xy}(x,y)=f_x(x) \cdot f_y(y)$
24	Two parameter of uniform distribution are	a and b
25	What is the correct formula of mean of hyper geometric distribution	$\frac{nk}{N}$
26	The experiment continuous until r successive are observed where r is specified in advance. This is an example of which distribution	Negative binomial distribution
27	What is the relationship between mean, median, mode of a normal distribution	Always same
28	How many parameter are in the Bivariate normal distribution	Five
29	The bivariate normal distribution can be expressed as	$N(\mu_x, \mu_y, \sigma_x^2, \sigma_y^2, \rho)$
30	The bivariate normal distribution has	Bell shape
31	In normal distribution, the all odd order moment of about mean are	Zero
32	In general, the expected value of a product is _____ to the product of the expected value that is $E(XY)$ _____ $E(X)E(Y)$	Not equal , $\neq$
33	Which distribution pass the quality that is means and standard deviation are equal	Exponential distribution
34	The row moment of the normal distribution is equal to the	Mean $E(X)$
35	All the higher cumulant of a normal distribution are equal to	Zero
36	In Bivariate Normal Distribution, what is the range limit of variable X and Y	$-\infty < x, y < \infty$
37	The first cumulant $K'(0)$ of normal distribution is	Mean (μ)
38	In the case of Bivariate normal distribution, the random variable X and Y are independent iff the correlation coefficient (ρ) =	$\rho = 0$
39	Choose the correct formula of VARIANCE of multinomial distribution	$np_i(1 - p_i)$
40	Consider the equation of circle $(x_1 - c_1)^2 + (x_2 - c_2)^2 = a^2$	a

MCQ BINOMIAL AND HYPERGEOMETRIC DISTRIBUTIONS

MCQ 8.1

A Bernoulli trial has:

- (a) At least two outcomes (b) At most two outcomes
(c) Two outcomes (d) Fewer than two outcomes

MCQ 8.2

The two mutually exclusive outcomes in a Bernoulli trial are usually called:

- (a) Success and failure** (b) Variable and constant
(c) Mean and variance (d) With and without replacement

MCQ 8.3

Nature of the binomial random variable X is:

- (a) Quantitative (b) Qualitative **(c) Discrete** (d) Continuous

MCQ 8.4

In a binomial probability distribution, the sum of probability of failure and probability of success is always:

- (a) Zero (b) Less than 0.5 (c) Greater than 0.5 **(d) One**

MCQ 8.5

In a binomial experiment, the successive trials are:

- (a) Dependent **(b) Independent** (c) Mutually exclusive (d) Fixed

MCQ 8.6

The parameters of the binomial distribution are:

- (a) n and p** (b) p and q (c) np and nq (d) np and npq

MCQ 8.7

The range of binomial distribution is:

- (a) 0 to n** (b) 0 to ∞ (c) -1 to +1 (d) 0 to 1

MCQ 8.8

The mean and standard deviation of the binomial probability distribution are respectively:

- (a) np and npq **(b) np and $\sqrt{npq}$** (c) np and nq (d) n and p

MCQ 8.9

In a binomial experiment with three trials, the variable can take:

- (a) 2 values (b) 3 values **(c) 4 values** (d) 5 values

MCQ 8.10

The shape of the binomial probability distribution depends upon the values of its:

- (a) Mean (b) Variance **(c) Parameters** (d) Quartiles

MCQ 8.11

In binomial distribution the numbers of trials are:

- (a) Very large (b) Very small **(c) Fixed** (d) Not fixed

MCQ 8.12

In a binomial probability distribution, relation between mean and variance is:

- (a) Mean < Variance (b) Mean = Variance
(c) Mean > Variance (d) Difficult to tell

MCQ 8.13

In binomial distribution when $n = 1$, then it becomes:

- (a) Hypergeometric distribution (b) Normal distribution
(c) Uniform distribution **(d) Bernoulli distribution**

MCQ 8.14

The mean of a binomial distribution depends on:

- (a) Number of trials (b) Probability of success
(c) Probability of failure **(d) Number of trials and probability of success**

MCQ 8.15

The variance of a binomial distribution depends on:

- (a) Number of trials (b) Probability of success
(c) Probability of failure **(d) All of the above**

MCQ 8.16

Which of the following is not property of a binomial experiment?

- (a) Probability of success remains constant
(b) n is fixed
(c) Successive trials are dependent
(d) It has two parameters

MCQ 8.17

The binomial probability distribution is symmetrical when:

- (a) $p = q$** (b) $p < q$ (c) $p > q$ (d) $np > npq$

MCQ 8.18

The binomial distribution is negatively skewed if:

- (a) $p < 1/2$ (b) $p = 1/2$ **(c) $p > 1/2$** (d) $p = 1$

MCQ 8.19

In a binomial probability distribution, the skewness is positive for:

- (a) $p < 1/2$** (b) $p = 1/4$ (c) $np = npq$ (d) $np = nq$

MCQ 8.20

Which of the following statements is false?

- (a) Expected value of a constant
(b) In a binomial distribution the standard deviation is always less than its variance
(c) In a binomial distribution the mean is always greater than its variance
(d) In binomial experiment the probability of success remains constant from trial to trial

MCQ 8.21

If a binomial probability distribution has parameters $(n, p) = (5, 0.6)$, the probability of $x = 3.5$ is:

- (a) 0** (b) 1 (c) 0.6 (d) 0.4

MCQ 8.22

In a binomial experiment $n = 4$, $P(x=2) = 216/625$ and $P(x=3) = 216/625$. $P(x=-2)$ is:

- (a) $216/625$ (b) 1 **(c) 0.6** (d) Difficult to tell

MCQ 8.23

If $n = 6$ and $p = 0.9$ then the value of $P(x=7)$ is:

- (a) Zero** (b) Less than zero (c) More than zero (d) One

MCQ 8.24

In a binomial probability distribution, coefficient of skewness = $(q - p)/\sqrt{npq} = 0$, it means that the distribution is:

- (a) Symmetrical** (b) Skewed to the left (c) Skewed to the right (d) Highly skewed

MCQ 8.25

For a binomial distribution with $n = 10$, $p = 0.5$, the probability of zero or more successes is:

- (a) 1** (b) 0.5 (c) 0.25 (d) 0.75

MCQ 8.26

In a binomial distribution, the mean, median and mode coincide when:

- (a) $p < 1/2$ (b) $p > 1/2$ (c) $p \neq 1/2$ **(d) $p = 1/2$**

MCQ 8.27

In which distribution, the probability success remains constant from trial to trial?

- (a) Hypergeometric distribution **(b) Binomial distribution**
(c) Sampling distribution (d) Frequency distribution

MCQ 8.28

In a binomial experiment when $n = 5$, the maximum number of successes will be:

- (a) 0 (b) 2.5 (c) 4 **(d) 5**

MCQ 8.29

In a binomial experiment when $n = 10$, the minimum number of successes will be:

- (a) 0** (b) 5 (c) 10 (d) 11

MCQ 8.30

If $n = 10$ and $p = 0.6$, then $P(x \geq 0)$ is:

- (a) 0.5 (b) 0.6 **(c) 1.0** (d) 1.2

MCQ 8.31

A random variable X has a binomial distribution with $n = 4$, the standard deviation of X is:

- (a) $4pq$ **(b) $2\sqrt{pq}$** (c) $4p$ (d) $4(q+p)$

MCQ 8.32

In a multiple choice test there are five possible answers to each of 20 questions. If a candidate guesses the correct answer each time, the mean number of correct answers is:

- (a) 4** (b) 5 (c) $1/5$ (d) 20

MCQ 8.33

If three coins are tossed, the probability of two heads is:

- (a) $1/8$ **(b) $3/8$** (c) $2/3$ (d) 0

MCQ 8.34

Random variable x has binomial distribution with $n = 8$ and $p = 1/2$. The most probable value of X is:

- (a) 2 (b) 3 **(c) 4** (d) 5

MCQ 8.35

The value of second moment about the mean in a binomial distribution is 36. The value of the standard deviation of a binomial distribution is:

- (a) 36 **(b) 6** (c) $1/36$ (d) $1/6$

MCQ 8.36

For a binomial probability distribution, the expected frequency of x successes in N experiments is:

- (a) $\frac{\binom{k}{x} \binom{N-k}{n-x}}{\binom{N}{n}}$ (b) $p^x(1-p)^{1-x}$
 (c) $N \binom{n}{x} p^x q^{n-x}$ (d) $\binom{n}{x} p^x(1-p)^{n-x}$

MCQ 8.37

In a binomial frequency distribution $100 (1/5 + 4/5)^5$. The parameters n and p are respectively:

- (a) (5, 1/5) (b) (1/5, 4/5) (c) (100, 4/5) **(d) (5, 4/5)**

MCQ 8.38

For a binomial frequency distribution $100 (1/5 + 4/5)^5$, the mean is:

- (a) 1/5 (b) 4/5 (c) 5 **(d) 4**

MCQ 8.39

For a binomial distribution $(1/3 + 2/3)^{18}$, the standard deviation of the binomial distribution will be:

- (a) 2** (b) 4 (c) 6 (d) 12

MCQ 8.40

The hypergeometric distribution has:

- (a) One parameter (b) Two parameters **(c) Three parameters** (d) Four parameters

MCQ 8.41

The parameters of the hypergeometric distribution are:

- (a) N, n, p (b) N, n, np **(c) N, n, k** (d) n and p

MCQ 8.42

Nature of the Hypergeometric random variable is:

- (a) Continuous **(b) Discrete** (c) Qualitative (d) Quantitative

MCQ 8.43

In hypergeometric distribution, the successive trials are:

- (a) Independent **(b) Dependent** (c) Very large (d) Very small

MCQ 8.44

In a hypergeometric distribution, the probability of success:

- (a) Remains constant from trial to trial
(b) Does not remain constant from trial to trial
 (c) Equal to probability of failure
 (d) Less than probability of failure

MCQ 8.45

If in a hypergeometric distribution $N = 10$, $k = 5$ and $n = 4$; then the probability of failure is:

- (a) 2 **(b) 0.5** (c) 1 (d) 0.25

MCQ 8.46

The rang of hypergeometric distribution is:

- (a) 0 to n (b) 0 to k (c) 0 to N **(d) 0 to n or k (whichever is less)**

MCQ 8.47

The number of trials in hypergeometric distribution is:

- (a) Not fixed **(b) Fixed** (c) Large (d) Small

MCQ 8.48

The probability of a success changes from trial to trial in:

- (a) Binomial distribution (b) **Hypergeometric distribution**
 (c) Normal distribution (d) Frequency distribution

MCQ 8.49

The mean of the hypergeometric distribution is:

- (a) $\frac{nk}{N}$ (b) $\frac{Nk}{n}$ (c) $\frac{Nn}{k}$ (d) $\frac{n+k}{N}$

MCQ 8.50

The standard deviation of the hypergeometric distribution is:

- (a) $\frac{nk(k-N)(N-n)}{N^2(N-1)}$ (b) $\sqrt{\frac{nk(N-k)(N-n)}{N^2(N-1)}}$
 (c) $\sqrt{\frac{nk(N-k)(n-N)}{N^2(N-1)}}$ (d) $\sqrt{\frac{nk(N-k) + (N-n)}{N^2(N-1)}}$

MCQ 8.51

In hypergeometric probability distribution, the relation between mean and variance is:

- (a) **Mean > variance** (b) Mean < Variance (c) Mean = Variance (d) Mean = 2Variance

MCQ 8.52

Which of the following is the property of hypergeometric experiment?

- (a) **p** remains constant from trial to trial
 (b) Successive trials are independent
 (c) **Sampling is performed without replacement**
 (d) **n** is not fixed

MCQ 8.53

Hypergeometric distribution reduces to binomial distribution when:

- (a) $N = n$ (b) $n \rightarrow \infty$ (c) **$N \rightarrow \infty$** (d) $N < n$

MCQ 8.54

In a hypergeometric distribution $N=6$, $n=4$ and $k=3$, then the mean is equal to:

- (a) **2** (b) 4 (c) 6 (d) 24

MCQ 8.55

Given $N = 11$, $n = 5$, $k = 7$; $P(x \geq 1)$ equals:

- (a) **1** (b) 1/66 (c) 65/66 (d) None of the above

MCQ 8.56

Given $N = 12$, $n = 5$, $k = 4$; $P(x \leq 4)$ equals:

- (a) Less than one (b) **Exactly one** (c) More than one (d) Between 0.5 and 1

1.(c)	2.(a)	3.(c)	4.(d)	5.(b)	6.(a)	7.(a)	8.(b)	9.(c)	10.(c)	11.(c)	12.(c)	13.(d)	14.(d)	15.(d)
16.(c)	17.(a)	18.(c)	19.(a)	20.(b)	21.(a)	22.(c)	23.(a)	24.(a)	25.(a)	26.(d)	27.(b)	28.(d)	29.(a)	30.(c)
31.(b)	32.(a)	33.(b)	34.(c)	35.(b)	36.(c)	37.(d)	38.(d)	39.(a)	40.(c)	41.(c)	42.(b)	43.(b)	44.(b)	45.(b)
46.(d)	47.(b)	48.(b)	49.(a)	50.(b)	51.(a)	52.(c)	53.(c)	54.(a)	55.(a)	56.(b)				

Sta 642 MCQ's:-

- ① Shape Parameters of beta distribution are.
 - (a) α, β (b) $-\alpha, \beta$ (c) α, β ✓
- ② Beta distribution defined on interval
 - (a) $[0, 1]$, $\alpha < 0$, $\beta < 0$ (b) ? (c) ?
 - ✓ (d) $[0, 1]$ $\alpha, \beta > 0$
- ③ When α increases in Beta distribution, the distribution become ?
 - (a) Normal (b) ? (c) ? (d) ?
- ④ Parameters of normal distribution are
 - ✓ (a) 1 (b) 2 (c) 4 (d) 5
- ⑤ Parameters of binomial distribution are
 - ✓ (a) n and p (b) $-p$ and n (c) ?
- ⑥ Variance of Hypergeometric distribution
 - ✓ (a) $\text{Var} = n \frac{K}{N} (1 - \frac{K}{N}) \frac{N-n}{N-1}$ (b) ? (c) ?
- ⑦ The variance of multinomial distn.
 - (a) $np_i(1-p_i)$ ✓ (b) $np_i(1-p_i)$
- ⑧ In which distribution the trials are ^{not} independent. **hyper**
 - (a) Binomial (b) Hypergeometric (c) Normal

9. The parameter of poisson dist. is
 (a) λ (b) $\frac{1}{\lambda}$ (c) λ^2 (d) ?
10. When we count the occurrence of an event with respect to time the distribution is
 (a) Binomial (b) Poisson (c) normal
11. Geometric distribution is the special case of —
 (a) Negative binomial distribution.
 (b) Binomial (c) ? (d) ?
12. The intervals $0 < x < y$ and $0 < y < x$ can be rewritten as —
 (a) $0 < x < 1, x < y < 1$ (b) vice versa
13. Correlation co-efficient is denoted as —
 (a) $CV(x, y)$ (b) $cov(x, y)$
14. Two random variables x and y are independent if and only if —
 (a) $F(x, y) = F_x(x) F_y(y)$
 (b) $F(x, y) = F_x(x) + F_y(y)$
 (c) $F(x, y) = F_x(x) - F_y(y)$
 (d) $F(x, y) = F_x(x) / F_y(y)$

(15) PDF stands for.

☒ Probability Density function

☐ Probability Distribution function.

(16) let the random variable X denotes the number of trials at which first success occurs. distn

☒ (a) Geometric (b) Negative binomial

☐ (c) Binomial distn —

(17) If $M(t)$ is a MGF then its derivatives

(a) $M'(t) = E(Xe^{mt}) = ?$

not sure

(18) Variance of continuous uniform distribution is —.

☒ (a) $\text{Var}(X) = \frac{(b-a)^2}{12}$ (b) $\frac{(a+b)^2}{12}$

(19) If two random variables are independent then $\text{COV}(X, Y) =$

(a) 1 ☒ (b) zero (c) ?

(20) $f(x) = \begin{cases} \text{some function?} & \text{limits is} \\ 0 & \text{elsewhere} \end{cases}$

☒ (a) Pdf (b) Pmf (c) CDF

F_X

(21) let X be $N(30, 50)$ then mean is

☒ (a) 20 (b) 30 (c) 80 (d) 50

(22) The mean, median and mode of normal distribution.

☒ (a) All are same ☐ (b) different

(23) For the normal distribution the odd order moments about the mean are —

☒ (a) All zero ☐ (b) 1 ☐ (c) 2

(24) The first raw moment is

☐ (a) zero ☐ (b) 1 ☒ (c) mean ☐ (d) variance

(25) The first cumulant of uniform dist is —

☐ (a) 0 ☒ (b) mean ☐ (c) 1

(26) All higher order cumulants —

☒ (a) zero ☐ (b) 1 ☐ (c) 2

(27) The joint Pdf of Bivariate normal distribution has shape.

☒ (a) bell shape ☐ (b) U shape ☐ (c) S shape

(28) The joint Probability normal dist referred to as

☒ (a) $N(\mu_x, \mu_y, \sigma_x^2, \sigma_y^2, \rho)$

☐ (b) $N(\mu_x, \mu_y, \sigma_x^2, \sigma_y^2, n)$

(29) The parameters of joint bivariate normal dist

MTWTFSS

DATE: _____

29. The parameters of joint ~~nor~~
bivariate normal distr — —

- (a) 2 (b) 1 (c) 4 (d) 5

30. The Pdf of normal distribution
is

(a) $f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$

31. In binomial distribution we
see the distribution with
parameters.

- (a) n and p (b) n and $-p$

32. $f(x) = \begin{cases} \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} e^{-x/\beta} & 0 < x < \infty, \alpha > 0, \beta > 0 \\ 0 & \text{elsewhere} \end{cases}$

is Gamma distribution.

- (a) Beta (b) Gamma (c) Normal

33. $\lambda e^{-\lambda x}$ is Exponential distribution

- (a) Poisson (b) Exponential

150. Even order moment $\sigma^{2n} = (2n-1)(2n-3) \dots 5 \cdot 3 \cdot 1 (\sigma^{2n})$

151. Standard normal Variance = σ^2

155. $f(x,y) = 2$ Conditional

49. $\lambda = 4$ CoVar = 50%

48. Ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ $(x_1 - c)^2 + (y_1 - c_2)^2 = a^2$

147. $y^2/3$ $f(x,y)$ given

46. Properties of Gamma

45. Co-eff of variance Poisson $\lambda = 1$

44. negative variance = $\frac{1-p}{p^2}$

43. Example Bivariate

42. $M(t) = (\frac{2}{3} + \frac{1}{3} e^t)^5$ $n=5$ $p = \frac{1}{3}$

141. $\theta = 1.5$

$$\frac{1}{\theta} \cdot e^{-\frac{1}{\theta}} = \frac{1}{1.5} e^{-\frac{1}{1.5}}$$

22. Not fixed in advance (H6)

21. Geo is special case of Binomial

20. Mean Binomial (mp)

19. $p = \frac{1}{2}$ Symmetrical

18. Uniform 2 parameter (a,b)

17. Shape parameter α, β (2)

Note: Submit this sheet to Superintendent, before leaving the Examination Center.

$$7 = (x+y) \leq (x_1+y_1)$$

6. $f(0) + f'(0)$
MacLaurin

5. Daily life (Degenerate)

4. Quartile (100 percent)

3. Inverse Cumulative

16. $0 \leq \alpha \leq 1$
 $\alpha \beta > 0$

15. α increase
Normality

$$14. \frac{1}{\Gamma(\alpha)\beta^\alpha} \lambda^{\alpha-1} e^{-\lambda/\beta}$$

Gamma

12. $\lambda e^{-\lambda x}$
pmf

12. $\frac{(a-b)^2}{12}$
variance of Cont.

11. Small period
C.D.F
Poisson
Normal

10. Mean Var
Poisson λ

9. $\frac{nK}{N}$ Hyper
 $n p$ mean
 $p = \frac{k}{N}$

8. not independent
Hyper

2. Mean Discrete Cont
1. Sample Space (Event)

40- First moment $m=1$ about origin
Mean

39- $N(30, 50)$ Mean = 30

38- Bivariate parameter n.p. ~~(X)~~ (S)

37- $(x_1 - c_1)^2 + (c_2 - c_1)^2 = a^2$ value of radius (a)

36- Independent $P=0$

35- Bivariate Normal $N(\mu_x, \mu_y, \sigma_x, \sigma_y, \rho)$

34- Bivariate normal (~~bell shaped~~) Rectangular

33- Higher cumulant = 0

32- first $K'(0) = \mu$

31- First raw moment Mean (u)

30- Mean = 0 odd

29- mean = med = mode Same Normal

28- Normal Dis (a, b) 2 parameter

27- $E(Z) = 1$ Stan Normal

26- St. Normal Special case of Normal

25- X, Y ind $F(X, Y) = F_X(x) F_Y(y)$

24- Domain $0 < x < y < 1$

$$\boxed{0 < x < y < 1}$$

23- Variance $n_i P_i(1 - P_i) \times$
mult $n P_i(1 - P_i) \checkmark$

$\frac{y-21}{5}$

ii For all $p \in P$, The value of $F_X(\infty)$

iii What is the expected value of the following distribution?

$$f(x) \propto e^{-\lambda x} \quad 0 < x < \infty, \lambda > 0$$

iv Central moment generating function is denoted by

v. Covariance is denoted by (6) Let $X = (X_1, X_2)$ be a random vector.

Then the expected value of X exists if the expectation(s) of

(7) One-to-one transformation, for two old variables, we get

8) The X and Y are equal in distribution in mathematically written as. (9) In which distribution the successive draws are not independent of each other (10) The Variance of the

Hypogeometric distribution is given by (11) Choose the correct parameter(s) of Poisson distribution.

(12) Which distribution has a unique property that its mean and variance are equal? (13) Which one the correct formula of

the variance of Continuum (14) For any fixed value of the scale parameter β as α increases, the shape of the distribution tends to (15) Limits of probability density function of the β distribution are (16) The shape parameters of beta

(17) In binomial distribution if the value of $p > \frac{1}{2}$ then the shape of the distribution (18) The experiment continues until

r successes are observed, where r is specified in advance?

(19) In which distribution, the number of trial are NOT fixed in advance. (20) X and Y are independent if and only if

(21) In normal distribution σ^2 denotes the

(22) For the standard normal distribution, $t(z)$ is equal to with which value?

(23) Choose the correct number of parameters for normal distribution.
24. What is relation between the mean, median and mode of a normal distribution? 25. Normal distribution the

all odd order moments about mean are 0. 26. The first raw

moment of the ND is equal to 27. The first cumulated

k' th of normal distribution is 28. All the higher cumulants

of a normal distribution are equal to 29. Bivariate

Normal distribution has 30 bivariate Normal distribution

can be expressed as:

31. How many parameters are in the Bivariate

Normal distribution?

Definition

A pair of random variables X_1 and X_2 have a **bivariate** normal distribution and they are referred to as **jointly normally distributed** random variables **if and only if** their joint probability density is given by

$$f(x_1, x_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} e^{-\left\{ \frac{1}{2(1-\rho^2)} \left(\frac{(x_1-\mu_1)^2}{\sigma_1^2} - 2\rho \left(\frac{x_1-\mu_1}{\sigma_1} \right) \left(\frac{x_2-\mu_2}{\sigma_2} \right) + \frac{(x_2-\mu_2)^2}{\sigma_2^2} \right) \right\}}$$

for $-\infty < x_1 < \infty$ and $-\infty < x_2 < \infty$,

where $\mu_1 \in \mathbb{R}$, $\mu_2 \in \mathbb{R}$, $\sigma_1 > 0$, $\sigma_2 > 0$, and $-1 \leq \rho \leq 1$.

This is referred to as the $N(\mu_x, \mu_y, \sigma_x^2, \sigma_y^2, \rho)$ distribution.

Quiz

This joint p.d.f., has a bell-type shape.

Bivariate Normal Distribution

with $\rho = 0$ and $\sigma_1 = \sigma_2 = 1$

$f(x_1, x_2)$
0.15
0.1



Q.3

- The standard normal distribution is a special case of the normal distribution .

← correlation

×

<

20 of 30

>

of the
Correlation Coefficient

4. Coefficient of **Correlation** possess the property of symmetry

$$\rho_{XY} = \rho_{YX}$$

5. Co-efficient of correlation measures only **linear correlation** between X and Y.

Hypergeometric Distribution

The following conditions characterize the hypergeometric distribution:

1. The result of each draw (the elements of the population being sampled) can be classified into one of two mutually exclusive categories (e.g. Pass/Fail or Employed/Unemployed).
2. The successive draws are **not independent** of each other.
3. The number of draws is **fixed** in advance.

Quiz

4. The probability of a success **changes** on each draw, as each draw decreases the population (*sampling without replacement* from a finite population).

Example:



The **variance** of the Multinomial distribution

$$\text{Var}(X_i) = np_i(1 - p_i)$$

The **covariance** of the Multinomial distribution

$$f''(x) = -\frac{1}{\sigma^3 \sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2} \left[1 - e^{-(x-\mu)^2/2\sigma^2} \frac{(x-\mu)^2}{\sigma^2} \right]$$

Substituting $x = \mu$ in $f''(x)$, we see that $f''(x) < 0$.

Quiz

Thus,

$x = \mu$ is the mode of the normal distribution.

Hence the median and mode are both equal to μ .

Since,

mean = median = mode,

the normal distribution is symmetrical and unimodal.



Quiz

In the special case $a = b$, equation (1) reduces to

$$\frac{(x_1 - c_1)^2}{a^2} + \frac{(x_2 - c_2)^2}{a^2} = 1 \Rightarrow (x_1 - c_1)^2 + (x_2 - c_2)^2 = a^2$$

and we **already** know that

$$(x_1 - c_1)^2 + (x_2 - c_2)^2 = a^2$$

is the equation of that particular circle with center at the point (c_1, c_2) and with radius a .

Hence, if we put $\sigma_1 = \sigma_2 = \sigma$ in equation (2), we obtain

$$\begin{aligned} \frac{(x_1 - \mu_1)^2}{k\sigma^2} + \frac{(x_2 - \mu_2)^2}{k\sigma^2} &= 1 \\ \Rightarrow (x_1 - \mu_1)^2 + (x_2 - \mu_2)^2 &= k\sigma^2 \quad \dots(3) \end{aligned}$$

which is the equation of a circle

with center at (μ_1, μ_2) and with radius $= k\sigma$.

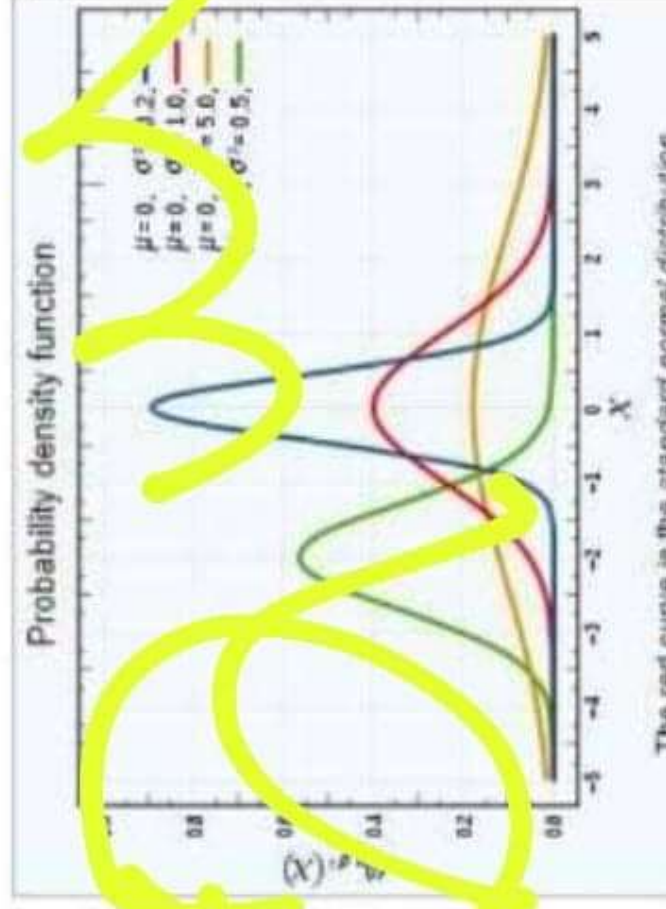
← null.pdf



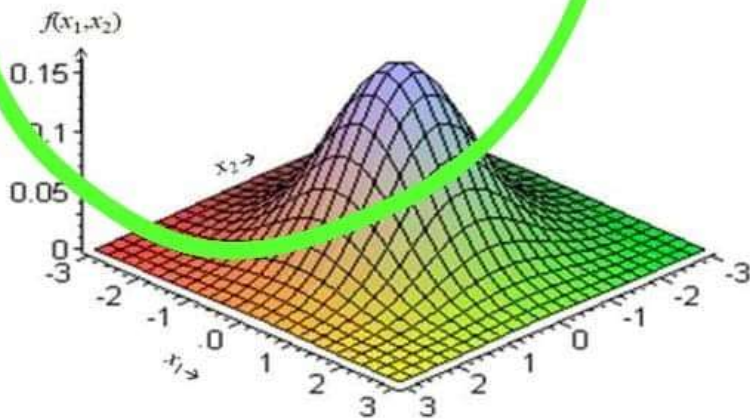
The graph points to the following **two basic properties** of the normal distribution:

1. The normal curve is

asymptotic to the x-axis,



The **circular contour** corresponding to a lower height is **wider** than the one corresponding to a greater height.



Next:

A circle can be regarded as

The First Three Cumulants

- The **first cumulant** is the **first moment about zero** i.e. the **mean** of the distribution.
- The **second cumulant** is the **second central moment** i.e. the **variance**.
- The **third cumulant** is the **third central moment**.



Higher Order Cumulants

The higher cumulants are neither moments about zero nor central moments, but rather **more complicated polynomial functions of moments**.

In fact, it can be shown that the k^{th} cumulant $\kappa_k(X)$ of a random variable X is the value of **a certain polynomial in the first k moments of X** i.e. a polynomial in $E(X^\ell)$, $\ell=1, \dots, k$

Derivation of the Mean and Variance of the Hypergeometric Distribution

Qwit

Two important properties of the hypergeometric probability distribution are given here.

The mean & variance of the Hypergeometric distribution are as follows:

$$\text{Mean} = E(X) = n \frac{K}{N}$$

&

$$\text{Variance} = n \frac{K}{N} \frac{(N-K)}{N} \frac{(N-n)}{N-1}$$

D5H

Because of the nature of the support of the bivariate density function, when finding **marginal** distributions:

Q ✓ the limits of the integral with respect to x are to be 0 to y
(and this will give us the marginal distribution of y i.e. $f_Y(y)$)

and

Q ✓ the limits of the integral with respect to y are to be x to 1.
(and this will give us the marginal distribution of x i.e. $f_X(x)$)

Quiz

• When $p = 1/2$, the distribution is always symmetrical.

In general:

• As n , the number of trials, increases to ∞ ,

$\beta_1 \rightarrow 0$ and $\beta_2 \rightarrow 3$.

✓ 507
Properties:

Ques 2

The mean of the Multinomial distribution

$$E(X_i) = np_i$$

The variance of the Multinomial distribution ✓

$$\text{Var}(X_i) = np_i(1 - p_i)$$

Q.12

Now, the mean μ and variance σ^2 of X may be computed from $M(t)$.

We know that $\mu = M'(0)$

Q.13

and

$$\sigma^2 = \underline{M''(0) - \mu^2}$$

Q. It is obvious that the negative binomial distribution is moderately positively skewed and that, as p tends to zero, the shape of the distribution tends to **normality**.

Two important properties of the hypergeometric probability distribution are given here.

The mean & variance of the Hypergeometric distribution are as follows: $\checkmark$ Mean = $E(X) = n \frac{K}{N}$

&

$$\checkmark \text{ Variance} = n \frac{K}{N} \left(\frac{N-K}{N} \right) \frac{N-n}{N-1}$$

p q

The variance of the Hypergeometric distribution are as follows:

✓

$$\text{Variance} = n \frac{K}{N} \frac{(N-K)}{N} \frac{(N-1)}{N-1}$$

QED

✓ Probability Distribution Function

The probability density function of the continuous uniform distribution is:


$$f(x) = \begin{cases} \frac{1}{b-a} \\ 0 \end{cases}$$

for $a \leq x \leq b$,

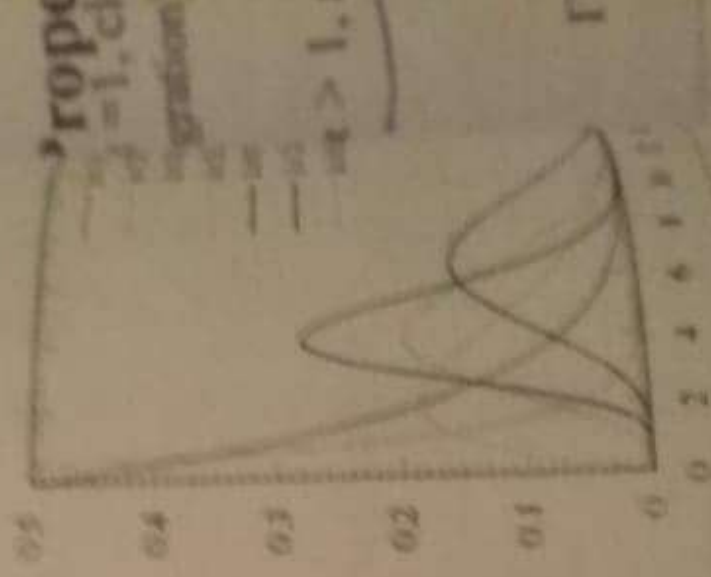
for $x < a$ or $x > b$

Shape of the distribution

From the graph it is obvious that, for any fixed value of the scale parameter β , as α increases, the shape of the distribution tends to normality.

Qm2

Probability density function



Gamma distribution:

Gamma distribution

$$f(x) = \begin{cases} \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} e^{-x/\beta} & 0 < x < \infty, \alpha > 0, \beta > 0 \\ 0 & \text{elsewhere} \end{cases}$$

is said to be the pdf of a gamma distribution with shape parameter α and scale parameter β and we write this as X has a $\Gamma(\alpha, \beta)$ distribution.

Qm

CDF of the Gamma Distribution

The CDF of the distribution is given by

$$F(x) = \begin{cases} \frac{1}{\Gamma(\alpha)} \gamma\left(\alpha, \frac{x}{\beta}\right) & \alpha > 0, \beta > 0 \\ 0 & \text{elsewhere} \end{cases}$$

where this $\gamma\left(\alpha, \frac{x}{\beta}\right)$ is the incomplete gamma function

pe of the C
can see that
er values of
parameter α , i.e.
and 1.0, the sh
e CDF appears
parabolic when
rger values of α
0, 3.0, 5.0 etc.,
is Skewed.

Mean:

$$\mu_X = E(X) = \underline{\alpha\beta}$$

Variance:

$$\sigma_X^2 = \underline{\alpha\beta^2}$$

Some Properties:

✓ • Mean $= \mu$

Qwr

$$= E(x) = \underline{a/a+\beta}$$

✓ • Variance(x)

$$= \underline{a\beta / (a+\beta)^2 (a+\beta+1)}$$

Q. 17

We know that, for any odd function g i.e. a function for which

$$g(-x) = -g(x) \quad \forall x \in \mathbb{R}$$

the integral $\int_{-\infty}^{\infty} g(x) dx$ is equal to zero.

Thus,

$x = \mu$ is the mode of the normal distribution.

Hence the median and mode are both equal to μ .

Since,

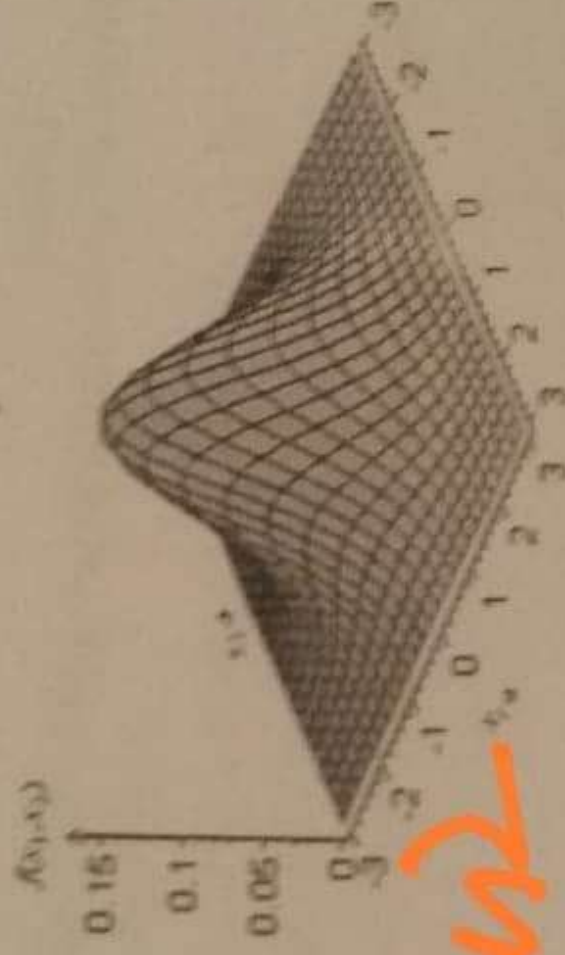
Qm 2

mean = median = mode,

the normal distribution is symmetrical and unimodal.

In other words, the points of inflection occur on the **right** and on the **left** of the mean **at a distance equal to standard deviation** and thus the graph of the normal curve is **bell-shaped**.

Bivariate Normal Distribution
with $\rho = 0$ and $\sigma_1 = \sigma_2 = \sigma$;



Quiz

An exact, beautiful bell-shaped surface

Hence we have the interesting property that the mean and standard deviation of the exponential distribution are equal.

This, in turn, means that the Coefficient of Variation of the Exponential distribution is always equal to 1.

In the special case $a = b$, equation (1) reduces to

$$\frac{(x_1 - c_1)^2}{a^2} + \frac{(x_2 - c_2)^2}{a^2} = 1 \Rightarrow (x_1 - c_1)^2 + (x_2 - c_2)^2 = a^2$$

and we already know that

$$(x_1 - c_1)^2 + (x_2 - c_2)^2 = a^2$$

is the equation of that particular circle with center at the point (c_1, c_2) and with radius a .

Quiz

Note that, in general, if X and Y are **independent** random variables, then their correlation coefficient is 0.

(In other words, it is ALWAYS true that in the case of two independent random variables X and Y , the correlation coefficient ρ is equal to **zero**.)

Quiz

Quiz

This is referred to as the $N(\mu_x, \mu_y, \sigma_x^2, \sigma_y^2, \rho)$ distribution.

$$\frac{1}{p} = \frac{1}{q} = \frac{1 \times 5}{1} = 2 \rightarrow$$

$$\frac{y}{x+B}$$

$$\frac{(x-y)^2}{(x+B)^2}$$

$$(x-y)^2$$

$$(x-y)^2 + (y -$$

$$\frac{70+100}{2}$$

$$\frac{100}{70}$$

$$\frac{170}{2} = 85$$

$$\frac{70}{100}$$

$$f(x) = x e^{-x} \left[\int_0^{\infty} t^{\alpha-1} e^{-t} dt \right]_{\alpha=1}$$

$$\int x dx$$

$$\frac{x}{x+B}$$

$$\frac{5 \times 1}{15+3} = \frac{16}{3}$$

For x

For y

$$\frac{x/B}{(x+B)^2}$$

$$\frac{16}{3} \times \frac{2}{1}$$

$$\frac{32}{3}$$

$$\int x dx$$

Note: Submit this sheet to Superintendent, before leaving the Examination Center.

Q

Hence we have the interesting property that the **mean and standard deviation** of the exponential distribution are equal.

Q

This, in turn, means that the **Coefficient of Variation of the Exponential distribution is always equal to 1.**

← null.pdf

Qwiz

Because of the nature of the support of the bivariate density function, when finding **marginal** distributions:

the limits of the **integral** with respect to **x** are to be **0 to y**
(and this will give us the marginal distribution of y i.e. $f_Y(y)$)

and

the limits of the **integral** with respect to **y** are to be **x to 1**.
(and this will give us the marginal distribution of x i.e. $f_X(x)$)

As such:

Marginal distribution of X :

$$f_X(x) = \int_x^1 2dy = 2.[y]_x^1 = 2(1-x), \quad 0 < x < 1$$

and

Marginal distribution of Y :

$$f_Y(y) = \int_0^y 2dx = 2.[x]_0^y = 2(y-0) = 2y, \quad 0 < y < 1$$

**Derivation of
the Mean and Variance of
the Hypergeometric Distribution**

Qwiz

Two important properties of the hypergeometric probability distribution are given here.

The mean & variance of the Hypergeometric distribution are as follows:

$$\text{Mean} = E(X) = n \frac{K}{N}$$



&

$$\text{Variance} = n \frac{K}{N} \frac{(N-K)}{N} \frac{(N-n)}{N-1}$$

STA642 Mega For Final Term March 2022

Prepared by Everblue

Subjective (14 files)

- Sta642 108 to 185 handwritten notes merge in 1 file pg81
- STA642 short Notes written by Mahar Afaq Safdar Muhammadi pg13
- M@l!k STA642 distribution notes pg33
- STA642 Spring 2021 paper solution Mahar Afaq Safdar Muhammadi pg31
- STA642 Current Subjective 2022 mark by Hussain Ali pg31
- STA642 Example By Mahar Afaq Safdar Muhammadi pg40
- STA642 Important Final Term By Moin Akhtar pg54
- Stat 642 Important final Question by Sir Sikandar pg60
- STA642 mega file for final term by Haseeb Sahib pg170
- STA642 Mega For Final Term All in 1 File 2022 pg308
- Sta642 current file 99% paper solved pg183
- STA642 Current quiz+paper 14to25March2022 pg130
- Sta642 latest final papers Rashid Hussain pg44
- STA642 Current Subjective 2022 mark by Hussain Ali pg31

Topic # 108

conditional distribution $\xrightarrow{\text{marginal distribution}}$ marginal distribution

limit $x \rightarrow 1$ $y \rightarrow 1$ m.g.f.

Example of computation of the variance of a conditional distribution.

Marginal distribution of X :

$$f_X(x) = \int_0^1 2 dy = 2 \cdot y \Big|_0^1 = 2 \cdot (1-0) = 2 \quad 0 < x < 1$$

and $\Rightarrow 2x - 2x = 0$
 $2 - 0 = 2$

Marginal distribution of Y :

$$f_Y(y) = \int_0^y 2 dx = 2 \cdot x \Big|_0^y = 2(y-0) = 2y \quad 0 < y < 1$$

For Conditional distributions

Formula: $\frac{\partial}{\partial y}$

(i) $f(x|y) = \frac{f(x,y)}{f_Y(y)} = \frac{2}{2y} = \frac{1}{y} \quad 0 < x < y$

and

(ii) $f(y|x) = \frac{f(x,y)}{f_X(x)} = \frac{2}{2(1-x)} = \frac{1}{1-x} \quad x < y < 1$

$\Rightarrow$ The conditional mean of X given $Y=y$

$E(X|Y=y) = \int_0^y x \cdot f(x|y) dx$

$\downarrow$ expected value

$$= \int_0^y x \cdot \frac{1}{y} dx = \frac{1}{y} \int_0^y x dx = \frac{1}{y} \left[\frac{x^2}{2} \right]_0^y = \frac{1}{y} \cdot \frac{y^2}{2} = \frac{y}{2} \quad 0 < y < 1$$

$$E(X|Y=y) = \frac{1}{y} \left[\frac{y^2}{2} \right] = \frac{y}{2} \quad 0 < y < 1$$

conditional mean

Imp

In case $U(x)$ is linear function of x

$$U(x) = (a + bx)$$

$E(X|y)$ is a linear conditional mean

M.C.Q

Topic No. 109

Now,

$$\begin{aligned} E(X^2|y) &= \int_0^y x^2 \cdot f(x|y) dx \Rightarrow \int_0^y x^2 \cdot \frac{1}{y} dx \\ &= \frac{1}{y} \int_0^y x^2 dx = \frac{1}{y} \left[\frac{x^3}{3} \right]_0^y \\ &= \frac{1}{y} \left[\frac{y^3}{3} - 0 \right] = \frac{y^2}{3}, \quad 0 < y < 1 \end{aligned}$$

M.C.Q

For Conditional Variance:

$$\begin{aligned} \text{Var}(X|y) &= E(X^2|y) - [E(X|y)]^2 \\ &= \frac{y^2}{3} - \left(\frac{y}{2}\right)^2 \Rightarrow \frac{y^2}{3} - \frac{y^2}{4} \\ &= \frac{4y^2 - 3y^2}{12} = \frac{y^2}{12}, \quad 0 < y < 1 \end{aligned}$$

Topic # 109

The Conditional Mean of Y given X that is linear in X .

$\Rightarrow$ Let $f(x, y)$ denote the joint Pdf of two random variable X and y
Let denote $f_1(x)$ the marginal Pdf of X
So,

$\Rightarrow$ The conditional Pdf of Y , given $X=x$

$$f_{2|1}(y|x) = \frac{f(x, y)}{f_1(x)}$$

at the points where $f_1(x) > 0$

The condition mean of Y , given $X=x$

$$E(Y|x) = \int_{-\infty}^{\infty} y f(y|x) dy$$

Expected value of random variable

$$= \int_{-\infty}^{\infty} \frac{y f(x, y)}{f_1(x)} dy = \int_{-\infty}^{\infty} y f(x, y) dy$$

(a function of x , say $u(x)$)

$\Rightarrow E(Y|X)$ is linear in X

$$E(Y|X) = \mu_2 + \rho \frac{\sigma_2}{\sigma_1} (X - \mu_1)$$

and

$$E(\text{Var}(Y|X)) = \sigma_2^2 (1 - \rho^2)$$

Actually,

$\Rightarrow$ both $E(Y|X)$ and $E(\text{Var}(Y|X))$ can be considered as random variables

نوٹ: چونکہ یہ X کی فکشن ہے اگر X کی فکشن کی فکشن ہے تو اسے $E(Y|X)$ کی فکشن بھی کہیں گے

Topic # 110

Independent Random Variables

Let X_1 and X_2 denote the random variable of the continuous type which joint Pdf $f(x_1, x_2)$.

By definition of conditional Pdf

$$f(x_1, x_2) = f_1(x_1) f_{2|1}(x_2|x_1)$$

If $f_{2|1}(x_2|x_1)$ does not depend upon x_1

$\Rightarrow$ the marginal variable Pdf of X_2

$$f_2(x_2) = \int_{-\infty}^{\infty} f(x_1, x_2) dx_1$$